

Coagulation disorder in the pregnancy

A stylized illustration of a blood vessel, possibly a vein, shown in cross-section. The vessel is red with an orange-yellow inner lining. Inside the vessel, there is a large, dark red, irregular mass representing a blood clot. Several small, red, disc-shaped objects (red blood cells) are scattered around the clot. A yellow, fibrous structure, likely a fibrin mesh, is shown trapping some of the red blood cells within the clot. The vessel continues to the right, narrowing slightly.

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Hematological changes in normal pregnancy

- Increase WBC
- Increase V,VII,VIII,IX,X,XII, fibrinogen, VWF
- Antithrombin III & protein C steady or decrease
- Decrease protein S,PAI(plasmenogen activator inhibitor)
- Platelet decrease(little change)

Venous Thromboembolism in Pregnancy:

Venous thromboembolism (VTE) refers to the formation of a thrombus within veins.

-This can occur anywhere in the venous system but the clinically predominant sites are:

in the vessels of the leg (giving rise to deep vein thrombosis, DVT) and in the lungs (resulting in a pulmonary embolus, PE).

Predisposing factors to VTE

- Activation of blood coagulation
 - Venous stasis
 - Endothelial injury
- (Virchow's triad)

Epidemiology

- In pregnancy risk of VTE increase to 1/1000 & is greatest in puerperum

Pregnancy is associated with 10 fold increase compared with the risk for non pregnant women

RISK factors for VTE & PTE

- 1.pre-existing
- Previous VTE
- Thrombophilia (antithrombin III deficiency, factor V Leiden, prothrombin gene variant, protein C or S deficiency)
- Anti-phospholipid syndrome
- Medical disorders (nephrotic syndrome, IBD)
- Sickle cell disease, gross varicose veins
- Age >35, BMI >30, parity >4

RISK factors for VTE & PTE

- 2-New onset or transient
- Surgical procedure in pregnancy or puerperum
- OHSS
- Hyperemesis, severe dehydration, blood loss
- Long haul travel, immobility >4days bed rest
- Pre-eclampsia
- Prolong labour, midcavity instrumental delivery
- Cesarean section, immobility after delivery
- Sever infection

Factors specific to pregnancy

- multiple pregnancy.**
- Pre-eclampsia .**
- Grand multiparity.**
- Instrument-assisted or Caesarean delivery.**
- Hemorrhage.**
- damage to the pelvic veins.**
- Prolonged labour.**
- Sepsis.**

Deep vein thrombosis:

Presentation :

- The most common symptoms are pain in the calf with varying degrees of redness or swelling.
- women's legs are often swollen in pregnancy but unilateral symptoms should give an alarm.
- Signs are few except that often the calf is tender to gentle touch.
- It is mandatory to ask about symptoms of PE as a woman with PE might present first with DVT.
- The difficulty is that some of these symptoms may be found in normal pregnancies.
- The patient may also be asymptomatic with a retrospective diagnosis being made following a PE.

Investigations and diagnosis

1.D-dimers:

- VTE is associated with increased levels of blood D-dimers and this is often used as a screening test in non-pregnant individuals.
- However levels of D- dimers are increased in uncomplicated pregnancy and levels increased with advancing pregnancy.
- A positive D- dimer screen is of no prognostic significance in VTE but *a negative D- dimer in pregnancy means no VTE (exclude VTE).*

2. Duplex ultrasound:

- This has a high sensitivity and specificity in proximal DVTs and is non-invasive.**
- It is unreliable for calf DVT as it has a much lower sensitivity.**
- If the initial ultrasound scan is negative and there is low level of clinical suspicion, anticoagulant treatment can be stopped.**
- If the ultrasound is negative and there is high clinical suspicion, the patient should be anti-coagulated and the ultrasound repeated in one week, or venography performed.**

3.Venography:

- Venography is invasive, requiring the injection of contrast medium and the use of x-ray.**
- it does allow excellent visualization of veins both below and above the knee.**
- But there is risks of radiation, allergic reaction and 5 percent risk of causing thrombosis.**

Pulmonary embolism- PE -

This can occur with or without preceding DVT.

-Symptoms range from minimal disturbance to sudden collapse and death, depending on the size, number and site of emboli.

Clinical features:

-It is crucial to recognize PE, as missing the diagnosis could have fatal implication.

The most common presentation is:

-mild dyspnea,

-inspiratory chest pain,

-tachycardia,

-mild pyrexia.

-Rarely massive PE may present with sudden cardiorespiratory collapse and even sudden death.

Investigation:

1-Arterial blood gas analysis- hypoxia and hypercapnia.

2-ECG: inverted T-wave and Q wave in lead III and atrial arrhythmias.

-In pregnancy T-inversion and Q-wave in lead III are normal findings

3-Chest x-ray: an abnormal CXR is found in 60-80% of patients with PE.

4-Ventilation-perfusion scan:

in cases of suspected PE, both V/Q scan and bilateral Doppler ultrasound of leg veins should be performed.

- Interpretation of a V/Q is given as probability rating .**
- Anticoagulation should be continued when the V/Q scan reports a medium or high probability of a PE .**
- If the scan reports a low probability and Doppler studies of the legs are positive anticoagulation should be continued.**
- If the Doppler is negative yet there is a high degree of clinical suspicion treatment continued with repeat testing after one week.**



5.Spiral CT : it can visualize the blood clot, also diagnose other diseases that mimic PE.

The radiation dose to the fetus is minimal.

6.Pulmonary angiogram:

The gold standard for the diagnosis of PE.

This is invasive, with mortality rate of 0.5% and associated morbidity of 2-4%.



management

-Medical anticoagulation is the treatment of choice for acute VTE.

-Heparin is the most frequently used drug, being non-toxic to the fetus (it does not cross the placental barrier).

-However, its main disadvantages are that it has to be parentally administered and on the long-term, may give rise to heparin-induced osteoporosis and thrombocytopenia.



-Warfarin

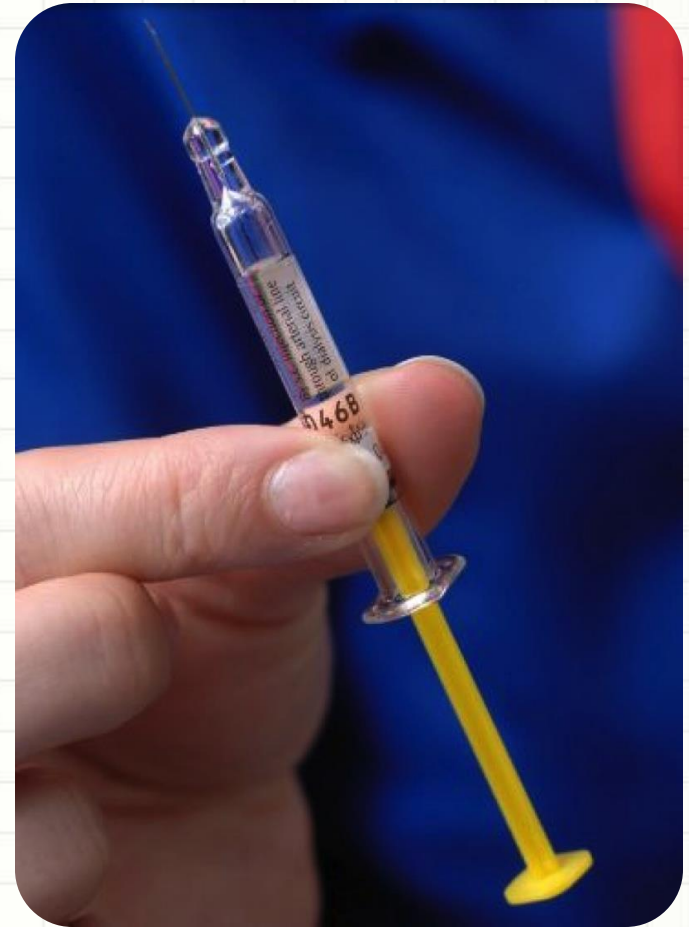
-use in the post-natal patient but it must be avoided during pregnancy as it is teratogenic (causing chondrodysplasia punctata in the 1st trimester) and can also cause placental abruption and fetal / neonatal hemorrhage in the 2nd and 3rd trimester .

-It act by inhibiting the synthesis of four vitamin K-dependent coagulant proteins (factors II, VII, IX, X) and at least two vitamin K-dependent anticoagulant factors, proteins C and S.

-It can be used safely during breast feeding.



- **In clinically suspected DVT or PE, treatment with low molecular weight heparin (LMWH) by subcutaneous route or unfractionated heparin should be given until the diagnosis is excluded by objective testing, unless treatment is strongly contraindicated.**



Initiating treatment

Baseline assessment:

- Carry out a full thrombophilia screen - this will not influence initial management but will provide information guiding the duration and intensity of long-term management.
- These results should be interpreted in the light of normal physiological changes during pregnancy.
- Check full blood count, coagulation screen, urea and electrolytes and liver function tests (renal and hepatic dysfunction will influence intensity of treatment).

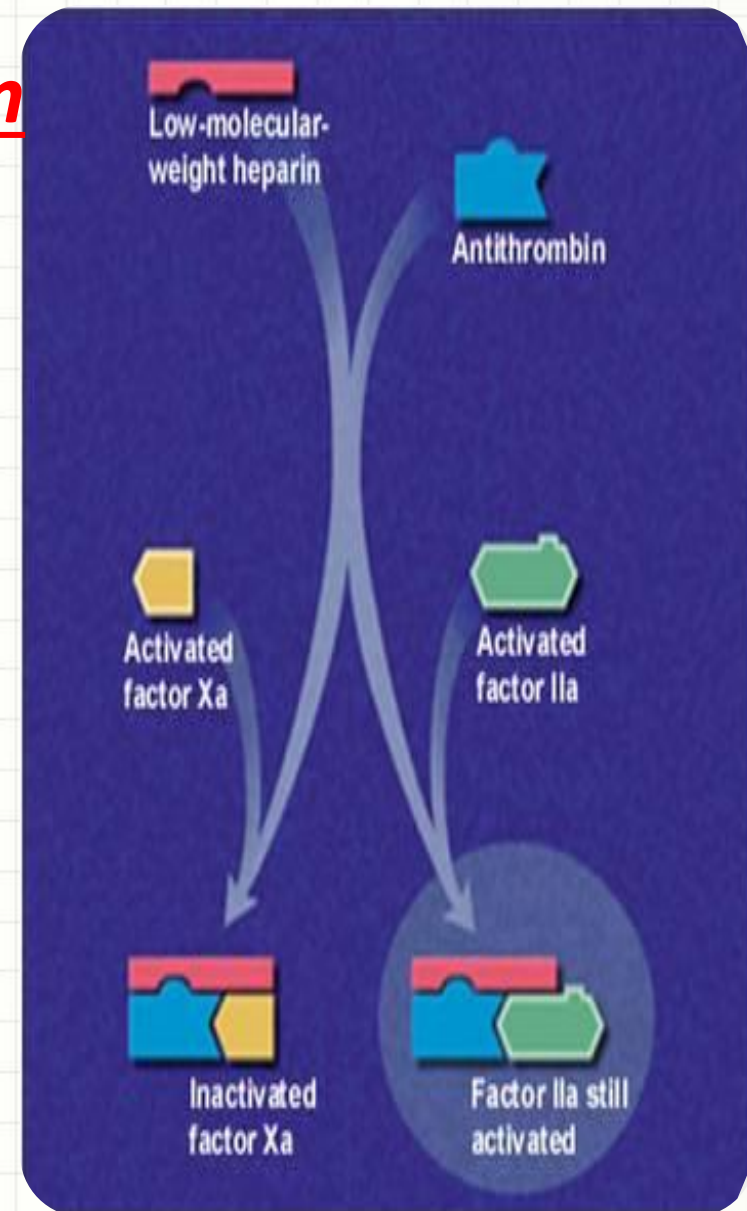
Low molecular weight heparin

- This has been shown to be more effective than unfractionated heparin with lower mortality and fewer hemorrhagic complications in the initial treatment of DVT in non-pregnant subjects.
- Also LMWHs are as effective as unfractionated heparin for treatment of PE.

LMWH

Subcutaneous

- Enoxaparin 1mg/kg twice daily
- Dalteparin 100 units/kg twice daily.



Maintenance therapy

During pregnancy

-Heparins are the maintenance treatment of choice

- Subcutaneous LMWH appears to have advantages over unfractionated heparin in the maintenance treatment of VTE in pregnancy (can be used once daily).**
- Women should be taught to self-inject and can then be managed as out-patients until delivery.**
- Unfractionated heparin (10000 units twice daily)**
- LMWH -Enoxaparin 40 mg daily**
-Dalteparin 5000 IU daily



Postpartum period

ongoing heparin treatment or Warfarin postpartum.

- If she opts for Warfarin, this can be initiated **day 2 or 3 post partum with an INR check at day 2.**
- Continue heparin treatment until there have been two successive readings of an INR > 2.
- It is not thought to pass into breast milk so it is safe during breast feeding.



Stopping treatment

Therapy is continued for **six** months in the first instance, as would be the case for non-pregnant patients.

- However, owing to the physiological fluctuation of coagulation factors, current advice is to continue therapy for at least 6-12 weeks post partum which ever the longer.
- At that point, the patient should be assessed for the presence of ongoing risk factors for a VTE prior to making the decision to stop anticoagulation therapy.

- **There is emerging evidence to suggest that compression stockings worn on the affected leg reduces the risk of developing post thrombotic syndrome and should be used for 2 years**



Prevention: prophylaxis

- There are obvious risks associated with ante-natal anticoagulation and the decision to go ahead with prophylactic thrombolysis is one made jointly by the obstetricians and hematologists.**
- Guidance suggested by the Royal College of Obstetricians and Gynecologists suggests**



- Regardless of their VTE risk, dehydration and immobilization of the patient ante-natally, during labour and post-partum should be avoided
- If a decision is made to go ahead with prophylaxis, this should be initiated as early in the pregnancy as possible (post-partum prophylaxis should commence as soon after the delivery as is practically possible)
- Women with a history of a VTE but no thrombophilia should be offered LMWH for 6 weeks post partum unless the VTE was clearly associated with a risk factor.



Antenatal Assessment & Management

(To be assessed at booking and repeated if admitted)

single previous VTE
+thrombophilia, or FH
Unprovoked/estrogen
related
Previous recurrent VTE



HIGH RISK
REQUIRES ANTENATAL
prophylaxis with LMWH

Single previous
provoked VTE
without FH of
thrombophilia
Thrombophilia +no
VTE



Intermediate risk require
close surveillance
antenatally &
prophylaxis with LMWH
for 6 W post partum

Antenatal Assessment & Management

(To be assessed at booking and repeated if admitted)

1. Age > 35 yrs
2. Obesity (BMI > 30 kg/m²)
3. Parity > 3
4. Smoker
5. MEDICAL CO-MORBIDITIES
e.g. heart or lung disease;
SLE; cancer; inflammatory conditions;
Proteinuria > 3g/24 hrs;
Sickle Cell Disease.
6. Gross varicose veins
7. Current systemic infection
8. Immobility, e.g. paraplegia,
SPD, long-haul travel
9. Pre-eclampsia
10. Dehydration/hyperemesis/OHSS
11. Multiple pregnancy or ART
12. Surgical procedure e.g. ERPC

3 or more risk
factor, 2 or
more if
admitted

**Intermediate
risk consider
antenatal
prophylaxis
with LMWH**

< 3 risk
factors

**Low risk
mobilisation
& avoidance
of dehydration**

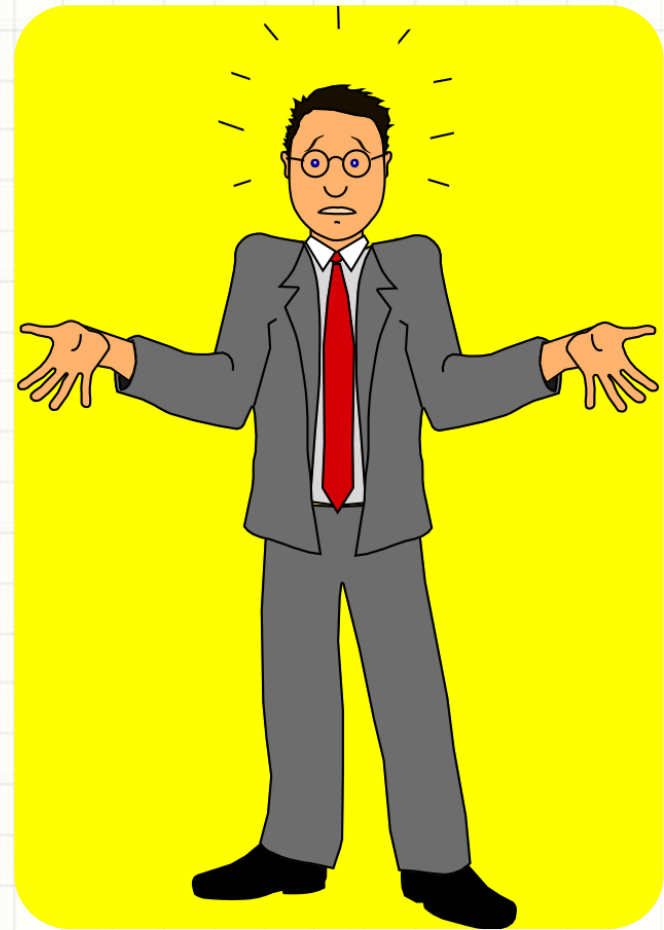
Timing of initiation of thromboprophylaxis



Timing of initiation of thromboprophylaxis

- **first trimester is associated with the greatest risk of VTE, many antenatal VTE events (including fatal events) occur in the first trimester**
- **So women should be advised to start LMWH as soon as they have a positive pregnancy test**

- which agent should be used for thromboprophylaxis



Which agents should be used for thromboprophylaxis?

- **LMWHs are the agents of choice for antenatal thromboprophylaxis. They are as effective and safer than unfractionated heparin.**

-Unfractionated heparin 10,000 IU/ per 12 hours

OR

-40 mg/day enoxaparin

OR

-Dalteparin 5000 IU/day S.C.

A close-up photograph of a green cactus with numerous sharp, light-colored spines. A large, vibrant red flower is in bloom on the right side of the cactus. The background is a clear, light blue sky.

Thank you